Bioavailability of orally administered water-dispersible hesperetin and its effect on peripheral vasodilatation in human subjects: implication of e...

Hesperetin is an aglycone of citrus flavonoids and is expected to exert a vasodilatation effect in vivo. We developed water-dispersible hesperetin by the process of micronization to enhance the bioavailability of hesperetin. This study aimed to assess the effect of this process on the bioavailability of hesperetin and to estimate its efficiency on vasodilatation-related functions using endothelial cells in vitro and a human volunteer study at a single dose in vivo. We found that water-dispersible hesperetin was absorbed rapidly, with its maximum plasma concentration (C(max)) being 10.2 ± 1.2 μM, and that the time to reach C(max), which is within 1 h if 150 mg of this preparation was orally administered in humans. LC-MS analyses of the plasma at C(max) demonstrated that hesperetin accumulated in the plasma as hesperetin 7-O-β-D-glucuronide (Hp7GA), hesperetin 3'-O-β-D-glucuronide (Hp3'GA) and hesperetin sulfate exclusively. Similar to hesperetin, Hp7GA enhanced nitric oxide (NO) release by inhibiting nicotinamide adenine dinucleotide phosphate-oxidase (NADPH oxidase) activity in a human umbilical vein endothelial cell culture system, indicating that plasma hesperetin metabolites can improve vasodilatation in the vascular system. A volunteer study using women with cold sensitivity showed that a single dose of water-dispersible hesperetin was effective on peripheral vasodilatation.These results strongly suggest that rapid accumulation with higher plasma concentration enables hesperetin to exert a potential vasodilatation effect by the endothelial action of its plasma metabolites. Water-dispersible hesperetin may be useful to improve the health effect of dietary hesperetin.